

* Soft Margin Classifier: [SVC]

$$\min_{w, w_0} \frac{1}{2} \|w\|_2^2$$

$$y^{(n)} (w^T x^{(n)} + w_0) \geq 1 - \epsilon^{(n)}$$

Epsilon $\rightarrow \epsilon^{(n)}$

slack variable

$$\sum_{n=1}^N \epsilon^{(n)} \leq C \quad \# \epsilon \geq 0$$

$\hookrightarrow$ hyperparameter

how many points are we allowing in the margin

or

but not preferred to use

$$\min_{w, w_0, \epsilon} \frac{1}{2} \|w\|_2^2 + C \sum_{i=1}^n \epsilon^{(n)}$$

penalty term

hard margin classifier $\downarrow$ null

side but since we have " $y^{(n)}$ " it always knows that the value is true.

if hard margin classifier is unable to plot a linear separable margin it will return null as in the above fig.

slack variable $\Rightarrow$ introduced in to ease the constraints

$\epsilon = \begin{cases} 0 & \rightarrow \text{Hard margin classifier} \\ [0, 1] & \rightarrow \text{inside the margin boundary or region} \\ > 1 & \rightarrow \text{other side of hyperplane} \end{cases}$

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* Primal Form

• Hard margin Classifier: (linear separability)

$$\min_{w, w_0} \frac{1}{2} \|w\|_2^2$$

such that,

$$y^{(n)} (w^T x^{(n)} + w_0) \geq 1 \quad \forall n$$

• Soft Margin: (linear separability)

$$(i) \min_{w, w_0, \epsilon} \frac{1}{2} \|w\|_2^2 + c \sum_{i=1}^n \epsilon^{(i)}$$

such that,

$$y^{(n)} (w^T x^{(n)} + w_0) \geq 1 - \epsilon^{(n)} \\ \epsilon^{(n)} \geq 0 \text{ for all } n$$

(ii)

$$\min_{w, w_0} \frac{1}{2} \|w\|_2^2$$

such that

$$y^{(n)} (w^T x^{(n)} + w_0) \geq 1 - \epsilon^{(n)} \\ \epsilon^{(n)} \geq 0; \sum_{i=1}^n \epsilon^{(i)} \leq c$$

in reality when we will use these, it is not exactly solved, like it's written & converted into

another mathematical form for mathematical computational efficiency, the older form Hard & soft margin are called primal form.

if c value is high, optimization function will try to reduce ϵ and $\frac{1}{2} \|w\|_2^2$ term.

* Dual Form or Lagrangian Form

Hard & Soft Margin Classifier are for linear data while SVM is for non-linear data.

* Support Vector Machine (SVM): (Non-linear separability)

$$\min_{w, w_0} \frac{1}{2} \|w\|_2^2$$

such that,

$$y^{(n)} (w^T x^{(n)} + w_0) \geq 1$$

$$\phi(x) \Rightarrow$$

it is non-linear we introduce kernel.

$$x \rightarrow \phi$$

ϕ = feature map

$$\min_{w, w_0} \frac{1}{2} \|w\|_2^2$$

This conversion

also works exactly

such that,

same for soft margin classifier.

primal form $\rightarrow y^{(n)} (w^T \phi(x^{(n)}) + w_0) \geq 1$

* The Dual Form:

2 values are feature map

dot product of 2 values in higher dimension space

$$\max_d \sum_{i=1}^n d^{(i)} - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n d^{(i)} d^{(j)} y^{(i)} y^{(j)} < \phi(x^{(i)}) \cdot \phi(x^{(j)})$$

$d \Rightarrow$ Lagrange Multiplier

replace with kernel functⁿ

subject to

$$d^{(i)} \geq 0$$

$$\sum d^{(i)} y^{(i)} = 0$$

$$K(x^{(i)}, x^{(j)})$$

There are five conditions which have to be satisfied to go from primal to dual form which are called KKT.

RBF Kernel

most popular kernel function in SVM.

Two hyperparameters to tune 'C' and 'gamma' for SVC, SVM.